

LeetCode Array Practice in C

15 beginner problems • Student Question Sheet

Instructions: Complete the missing logic inside each main(). Use only basic C concepts: arrays, loops, if/else, scanf and printf.

Do not worry about LeetCode function syntax yet. These are practice versions of the problems.

Problem List

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1. LeetCode 1480 — Running Sum of 1d Array

Problem: Given an array of integers, make a new value at every position equal to the sum of all elements from the beginning up to that position.

Sample:

Input: n = 4, arr = 1 2 3 4

Output: 1 3 6 10

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

2. LeetCode 1470 — Shuffle the Array

Problem: You are given $2n$ numbers arranged as $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$. Print them as $x_1, y_1, x_2, y_2, \dots, x_n, y_n$.

Sample:

Input: $n = 3$, $\text{arr} = 2\ 5\ 1\ 3\ 4\ 7$

Output: $2\ 3\ 5\ 4\ 1\ 7$

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100], ans[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

3. LeetCode 1431 — Kids With the Greatest Number of Candies

Problem: For every child, check whether giving them all the extra candies makes their number of candies at least as large as the maximum number any child currently has.

Sample:

Input: n = 5, candies = 2 3 5 1 3, extra = 3

Output: true true true false true

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100], extra;

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

4. LeetCode 1672 — Richest Customer Wealth

Problem: Each row represents one customer and contains the money in their bank accounts. Find the largest total wealth of any customer.

Sample:

Input: 2 customers, 3 accounts each

1 2 3

3 2 1

Output: 6

Start with this template:

```
#include <stdio.h>

int main() {
    int rows, cols, a[20][20];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

5. LeetCode 1920 — Build Array from Permutation

Problem: For every index i , create a new value using $\text{arr}[\text{arr}[i]]$. Print the resulting array.

Sample:

Input: 0 2 1 5 3 4
Output: 0 1 2 4 5 3

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100], ans[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

6. LeetCode 1365 — How Many Numbers Are Smaller Than the Current Number

Problem: For every element, count how many other elements in the array are strictly smaller than it.

Sample:

Input: 8 1 2 2 3

Output: 4 0 1 1 3

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100], ans[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

7. LeetCode 1295 — Find Numbers with Even Number of Digits

Problem: Count how many numbers in the array contain an even number of digits.

Sample:

Input: 12 345 2 6 7896

Output: 2

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

8. LeetCode 1512 — Number of Good Pairs

Problem: A good pair is a pair of positions i and j where $i < j$ and $arr[i]$ is equal to $arr[j]$. Count all good pairs.

Sample:

Input: 1 2 3 1 1 3

Output: 4

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

9. LeetCode 2006 — Count Number of Pairs With Absolute Difference K

Problem: Count pairs of positions i and j where $i < j$ and the absolute difference between their values is exactly k .

Sample:

Input: 1 2 2 1, $k = 1$

Output: 4

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100], k;

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

10. LeetCode 1 — Two Sum

Problem: Find two different positions whose values add up to the given target. Print their indexes.

Sample:

Input: arr = 2 7 11 15, target = 9

Output: 0 1

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100], target;

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

11. LeetCode 217 — Contains Duplicate

Problem: Check whether any number appears at least twice in the array.

Sample:

Input: 1 2 3 1

Output: true

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

12. LeetCode 27 — Remove Element

Problem: Remove every occurrence of a given value. Print the remaining elements and their count.

Sample:

Input: arr = 3 2 2 3, val = 3

Output: count = 2, array = 2 2

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100], val;

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

13. LeetCode 283 — Move Zeroes

Problem: Move all zero values to the end of the array while keeping the order of the non-zero values.

Sample:

Input: 0 1 0 3 12

Output: 1 3 12 0 0

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

14. LeetCode 268 — Missing Number

Problem: The array contains different numbers from 0 to n, but exactly one number is missing. Find the missing number.

Sample:

Input: 3 0 1

Output: 2

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.

15. LeetCode 136 — Single Number

Problem: Every number appears twice except one number. Find the number that appears only once.

Sample:

Input: 4 1 2 1 2

Output: 4

Start with this template:

```
#include <stdio.h>

int main() {
    int n, arr[100];

    // write your code here

    return 0;
}
```

Try to complete the program without looking at the solution.